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Latency function estimation under the mixture cure model when the cure status is available

Wende Clarence Safari[1] · **Ignacio López-de-Ullibarri**[2] · **María Amalia Jácome**[3]

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Abstract
This paper addresses the problem of estimating the conditional survival function of the lifetime of the subjects experiencing the event (latency) in the mixture cure model when the cure status information is partially available. The approach of past work relies on the assumption that long-term survivors are unidentifiable because of right censoring. However, in some cases this assumption is invalid since some subjects are known to be cured, e.g., when a medical test ascertains that a disease has entirely disappeared after treatment. We propose a latency estimator that extends the nonparametric estimator studied in López-Cheda et al. (TEST 26(2):353–376, 2017b) to the case when the cure status is partially available. We establish the asymptotic normality distribution of the estimator, and illustrate its performance in a simulation study. Finally, the estimator is applied to a medical dataset to study the length of hospital stay of COVID-19 patients requiring intensive care.

Keywords Bootstrap bandwidth · Censoring · Cure model · COVID-19 · Nadaraya-Watson weights

1 Introduction

When analyzing times to event in survival analysis, it is commonly assumed that the event of interest will happen when there is a sufficient follow-up time. However, there

✉ Wende Clarence Safari
wende.safari@lshtm.ac.uk

1 Inequalities in Cancer Outcomes Network (ICON), Department of Non-Communicable Disease Epidemiology, Faculty of Epidemiology and Population Health, London School of Hygiene and Tropical Medicine, London, UK

2 Department of Mathematics, Escuela Universitaria Politécnica, University of A Coruña, Ferrol, Spain

3 Department of Mathematics, Faculty of Science, University of A Coruña, CITIC, A Coruña, Spain

are many instances where the event will not occur for all the individuals. Some cancer patients will never relapse or die from cancer, some bank customers will never default in loan repayment, etc. Those whose event is certain not to occur are considered "statistically cured" (or long-term survivors) and those who will experience the event are known to be "uncured" (or susceptible) subjects. Cure models (Peng and Yu 2021) have been developed to address this issue. In this work we consider the well-known mixture cure model (MCM) (Boag 1949), where the probability to be cured and the distribution of the time to event for the uncured subjects (latency) are separately modeled. Modeling the distribution of the time to event in presence of cured individuals with classical estimators for the survival function is inappropriate, and tailored estimators of the latency function should be considered instead. Several authors have proposed methods to estimate the latency function in MCM, from either a (semi)parametric point of view (Amico and Van Keilegom 2018; Maller and Zhou 1992; Patilea and Van Keilegom 2020, among others), or from a completely nonparametric approach (López-Cheda et al. 2017b).

A difficulty encountered in the standard cure models is that the long-term survivors are never observed to be cured, rather they are censored at the end of the study. Hence, the cure status is unobserved in the right-censored subjects, and it is usually considered a latent variable in the analysis of survival data with a cured fraction. Nonetheless, in a number of situations, the cure status is observed for some censored individuals, as they can be identified as cured from the event. For example, diagnostic procedures in medical studies are available to provide further information on whether a subject can be considered as cured or not. Also, for some types of cancer it is extremely unlikely to have any recurrence later than a given time after treatment, known as cure threshold, and consequently those patients with observed time surpassing the cure threshold can be considered as cured from the recurrence (Taylor 1995). Our motivating example comes from a study of COVID-19 patients hospitalized in Galicia (North West Spain) during the outbreak of the pandemic. We had access to a medical database of 10,454 COVID-19 patients reported by the Galician Healthcare Service between March 6 and May 7, 2020. This database contains information on sex, age, and dates of clinical outcomes like admission to the intensive care unit (ICU), discharge or death. The aim was to estimate the time from hospital ward until admission to ICU, adjusting for age and sex. Thus, the event of interest was admission to ICU from hospital ward. Note that in this example "being cured from the event" means being free of experiencing admission to ICU, and not being cured in medical terms. In our analysis we included only 2,380 patients who had been hospitalized for at least one day. Among them, 8.3% were admitted to ICU and 91.7% were censored. In the censored group, 68.8% patients were discharged alive from the hospital without requiring ICU, and 13.8% died before ICU admission. Therefore, these two groups of patients will never enter ICU, so they can be identified to be "cured" from the event of interest, which is admission to ICU. Full details on the data are given in Sect. 4.

In this type of examples with a subgroup identified as cured from the event, there are three groups of observations: the event times, the "cured" times (of those acknowledged as cured from the event), and "regular" censored times (those who neither experienced the event by the end of the study nor classified as cured). Failure time data with a subgroup of individuals identified as cured from the event has become

increasingly frequent. Most of the methods proposed in the literature to estimate the survival function of the time to event (latency) consider a (semi)parametric model to handle covariates: Wu et al. (2014) modeled the latency part by the Cox proportional hazards model; Bernhardt (2016) studied the MCM when the latency function is modeled using a semiparametric accelerated failure time (AFT) model with a potentially known cure threshold. The estimation of the latency function using a completely nonparametric estimator, in the spirit of the Kaplan-Meier estimator, has been addressed only by Laska and Meisner (1992) by defining cure as survival beyond a fixed time u called *cure threshold*, and Betensky and Schoenfeld (2001) whose method allows for the observed "cured" times to be random. Neither of these nonparametric proposals can handle covariates.

We propose a fully nonparametric estimator of the conditional latency function. Our method extends the proposals in Laska and Meisner (1992) and Betensky and Schoenfeld (2001) to handle covariates, and the conditional nonparametric latency estimator in López-Cheda et al. (2017b) to incorporate the cure status information. The paper proceeds as follows. In Sect. 2, after describing the model notation, a new nonparametric kernel estimator of the conditional latency function is proposed. Also, the asymptotic properties of the estimator are studied and a bootstrap bandwidth selector is proposed. In Sect. 3, the results of a simulation study carried out to evaluate the finite sample performance of the estimator are presented. An application to the COVID-19 data is shown in Sect. 4. We conclude with a brief discussion in Sect. 5. The proofs and additional simulations are provided in Supplementary Material.

2 Estimation when the cure status is partially available

2.1 Model notation

Let Y be the time until the event of interest, $\mathbf{X}$ a vector of covariates and $F(t \mid \mathbf{x}) = P(Y \leq t \mid \mathbf{X} = \mathbf{x})$ the distribution function of Y conditional on $\mathbf{X} = \mathbf{x}$. In follow-up studies, the event of interest may not be observed due to, for example, the end of the study or loss to follow-up, which occur at the censoring time C with conditional distribution function $G(t \mid \mathbf{x}) = P(C \leq t \mid \mathbf{X} = \mathbf{x})$. As a consequence, instead of observing Y, only the minimum of Y and C time, and the indicator of the event $\delta = \mathbf{1}(Y < C)$ can be observed. The random variables Y and C are assumed to be conditionally independent given $\mathbf{X} = \mathbf{x}$, which is a widely used assumption in the literature. We set $Y = \infty$ if the subject will not experience the event, so is cured. Let $\nu = \mathbf{1}(Y = \infty)$ be the indicator of being cured. Note that ν is partially observed because when the event happened ($\delta = 1$) the individual is known not to be cured ($\nu = 0$), but in the general situation ν is unknown when $\delta = 0$. When the cure status is partially known, some censored individuals are identified to be cured, so $\nu = 1$ is observed.

To accommodate the possible availability of the cure status information, we include an additional random variable ξ, which indicates whether the cure status ν is known ($\xi = 1$) or not ($\xi = 0$). Further, let the censoring distribution be an improper distribution function $G(t \mid \mathbf{x}) = \{1 - \pi(\mathbf{x})\} G_0(t \mid \mathbf{x})$, so with probability $\pi(\mathbf{x})$ the

censoring variable is $C = \infty$, and with probability $1 - \pi(\mathbf{x})$ the value of the censoring variable C corresponds to the value of a random variable C_0 with proper continuous distribution function $G_0(t \mid \mathbf{x})$. When Y and C are conditionally independent given $\mathbf{X}$, then a cured individual is identified with probability $P(\xi = 1 \mid \nu = 1, \mathbf{X} = \mathbf{x}) = P(C = \infty \mid \mathbf{X} = \mathbf{x}) = \pi(\mathbf{x})$. The observed data $\{(\mathbf{X}_i, T_i, \delta_i, \xi_i \nu_i) : i = 1, \ldots, n\}$ can be classified into three groups: (a) the individual is observed to have experienced the event and therefore known to be uncured $(\mathbf{X}_i, T_i, \delta_i = 1, \xi_i \nu_i = 0)$; (b) the lifetime is censored and the cure status is unknown $(\mathbf{X}_i, T_i, \delta_i = 0, \xi_i \nu_i = 0)$; and (c) the individual is known to be cured $(\mathbf{X}_i, T_i, \delta_i = 0, \xi_i \nu_i = 1)$, where $T_i = \min(Y_i, C_i)[1 - \mathbf{1}(Y_i = \infty, C_i = \infty)] + C_{0i}\mathbf{1}(Y_i = \infty, C_i = \infty)$. In standard cure models when the cure status is unknown for all the censored observations $(\pi(\mathbf{x}) = 0, \xi_i \nu_i = 0$ for $i = 1, \ldots, n)$, only groups (a) and (b) are considered.

The probability of cure is $1 - p(\mathbf{x}) = P(Y = \infty \mid \mathbf{X} = \mathbf{x})$, and the conditional survival function of the uncured individuals, also known as latency, is $S_0(t \mid \mathbf{x}) = P(Y > t \mid Y < \infty, \mathbf{X} = \mathbf{x})$. The mixture cure model specifies the survival function $S(t \mid \mathbf{x}) = P(Y > t \mid \mathbf{X} = \mathbf{x})$ as

$$S(t \mid \mathbf{x}) = 1 - p(\mathbf{x}) + p(\mathbf{x}) S_0(t \mid \mathbf{x}). \tag{1}$$

One key issue in cure models is identifiability. Hanin and Huang (2014), and references therein, have discussed the condition needed for the MCM to be identifiable. They demonstrated that $\lim_{t \to \infty} S_0(t \mid \mathbf{x}) = 0$ for all $\mathbf{x}$ is a sufficient condition for identifiability of model (1). In this paper, we assume this condition holds.

2.2 Proposed estimator

Without loss of generality, let us consider a continuous covariate X with density function $m(x)$. By (1), the latency function can be written in terms of the survival function and cure probability as follows:

$$S_0(t \mid x) = \frac{S(t \mid x) - \{1 - p(x)\}}{p(x)}. \tag{2}$$

Following the same ideas as in López-Cheda et al. (2017b), we use the relation (2) to motivate our proposed nonparametric estimator of the latency function, replacing $S(t \mid x)$ and $1 - p(x)$ with efficient estimators. Safari et al. (2021) proposed the generalized product-limit estimator of the conditional survival function $S(t \mid x)$ when the cure status is partially known, which is

$$\widehat{S}_h^c(t \mid x) = \prod_{i=1}^{n} \left\{ 1 - \frac{\delta_{[i]} B_{h[i]}(x) \mathbf{1}\left(T_{(i)} \leq t\right)}{\sum_{j=i}^{n} B_{h[j]}(x) + \sum_{j=1}^{i-1} B_{h[j]}(x) \mathbf{1}\left(\xi_{[j]} \nu_{[j]} = 1\right)} \right\}, \tag{3}$$

where $X_{[i]}$, $\delta_{[i]}$, $\xi_{[i]}$ and $\nu_{[i]}$ are the concomitants of the ordered observed times $T_{(1)} \leq \cdots \leq T_{(n)}$, $B_{h[i]}(x)$ are the Nadaraya-Watson weights (Nadaraya 1964; Watson 1964),

$$B_{h[i]}(x) = \frac{K_h\left(x - X_{[i]}\right)}{\sum_{j=1}^{n} K_h\left(x - X_j\right)}, \tag{4}$$

and $K_h(\cdot) = K(\cdot/h)/h$ is a kernel function $K(\cdot)$ rescaled with bandwidth h. The corresponding estimator of the cure probability $1 - p(x)$ (Safari et al. 2022) is

$$1 - \widehat{p}_h^c(x) = \widehat{S}_h^c\left(T_{(n)}^1 \mid x\right), \tag{5}$$

where $T_{(n)}^1$ is the largest uncensored observed time. The optimal bandwidth for $\widehat{S}_h^c(t \mid x)$ in (3) is not necessarily the optimal bandwidth for $1-\widehat{p}_h^c(x)$ in (5). Therefore, we propose to estimate the latency function as follows:

$$\widehat{S}_{0,h_1,h_2}^c(t \mid x) = \begin{cases} \dfrac{\widehat{S}_{h_2}^c(t \mid x) - \{1 - \widehat{p}_{h_1}^c(x)\}}{\widehat{p}_{h_1}^c(x)} & \text{if } 0 \leq t \leq T_{(n)}^1 \text{ and } \widehat{S}_{h_2}^c(t \mid x) > 1 - \widehat{p}_{h_1}^c(x) \\ 0 & \text{otherwise,} \end{cases} \tag{6}$$

which is a proper non-increasing survival function. In the unconditional case, the estimator in (6) becomes

$$\widehat{S}_{0,n}^c(t) = \frac{\widehat{S}_n^c(t) - (1 - \widehat{p}_n^c)}{\widehat{p}_n^c}, \tag{7}$$

where

$$\widehat{S}_n^c(t) = \prod_{i=1}^{n} \left\{ 1 - \frac{\delta_{[i]}\mathbf{1}\left(T_{(i)} \leq t\right)}{n - i + 1 + \sum_{j=1}^{i-1} \mathbf{1}\left(\xi_{[j]}\nu_{[j]} = 1\right)} \right\}$$

is the generalization of the Kaplan-Meier estimator of the survival function to the presence of cured individuals with some of them identified as cured proposed by Safari et al. (2021), and

$$1 - \widehat{p}_n^c = \widehat{S}_n^c\left(T_{(n)}^1\right)$$

is the unconditional estimator of the probability of cure by Safari et al. (2022). Both estimators extend the method in Laska and Meisner (1992) for randomly observed "cured" times.

It is worth mentioning that the MCM when the cure status is partially known can be tailored in a competing risks context by considering the "Event of interest" and "Cure" as the competing events. The main disadvantage of the competing risks approach is that there is no a unique estimator of the latency function if the last observation is censored (neither an event nor a cure), only lower and upper bounds can be given. The

difference between both bounds increases with the censoring rate, see Betensky and Schoenfeld (2001) and Safari et al. (2022).

2.3 Asymptotic results

We will establish the asymptotic properties of the estimator in (6). We list the assumptions we make in obtaining these results in Appendix A in the Supplementary Material. Assumptions 1–9 are basically the same as in Safari et al. (2021, 2022) and have been commonly used in the literature (Iglesias-Pérez and González-Manteiga 1999; López-Cheda et al. 2017a, b). Additionally, we consider Assumption 10 to prove the asymptotic normality of the proposed estimator $\widehat{S}^c_{0,h_1,h_2}(t \mid x)$. Let us begin by defining some (sub)distribution functions:

$$\begin{aligned} H(t \mid x) &= P(T \leq t \mid X = x), \\ H^1(t \mid x) &= P(T \leq t, \delta = 1 \mid X = x), \\ H^{11}(t \mid x) &= P(T \leq t, \xi = 1, \nu = 1 \mid X = x), \\ J(t \mid x) &= 1 - H(t \mid x) + H^{11}(t \mid x). \end{aligned}$$

Finally, we require a further condition that is needed when estimating nonparametric cure models (López-Cheda et al. 2017a, b; Xu and Peng 2014). Let us define $\tau_H(x) = \inf\{t : H(t \mid x) = 1\}$, $\tau_0(x) = \inf\{t : S_0(t \mid x) = 0\}$ and $\tau_{G_0}(x) = \inf\{t : G_0(t \mid x) = 1\}$. Note that $\tau_H(x) = \max\left\{\tau_0(x), \tau_{G_0}(x)\right\}$ since $T = C_0$ if $Y = \infty$ and $T = \min(Y, C)$ otherwise. Let $\tau_0 = \sup_{x \in I} \tau_0(x)$, where I is defined in Assumption 1 of the Supplementary Material, then we require:

$$\tau_0 < \tau_{G_0}(x) \text{ for any } x \text{ with probability 1.} \tag{8}$$

Condition (8) ensures that uncured subjects will experience the event within the follow-up period, which entails that the estimator $1 - \widehat{p}^c_{h_1}(x)$ does not overestimate the cure probability.

Theorem 1 (Asymptotic representation) *Suppose that Assumptions 1–8 hold, and the bandwidth sequences $h_j = (h_{jn})$, for $j = 1, 2$, satisfy Assumption 9 as $n \to \infty$. Then, for $x \in I$, and $t \in [a, b]$ such that $\widehat{S}^c_{h_2}(t \mid x) > 1 - \widehat{p}^c_{h_1}(x)$, we have*

$$\widehat{S}^c_{0,h_1,h_2}(t \mid x) - S_0(t \mid x) = \sum_{i=1}^{n} \eta_{h_1,h_2}(T_i, \delta_i, \xi_i, \nu_i, t, x) + R_n(t, x) \tag{9}$$

where

$$\begin{aligned} \eta_{h_1,h_2}(T_i, \delta_i, \xi_i, \nu_i, t, x) = &- \frac{S(t \mid x)}{p(x)} \widetilde{B}_{h_2 i}(x) \zeta(T_i, \delta_i, \xi_i, \nu_i, t, x) \\ &- \frac{\{1 - p(x)\}\{1 - S(t \mid x)\}}{p^2(x)} \widetilde{B}_{h_1 i}(x) \zeta(T_i, \delta_i, \xi_i, \nu_i, \tau_0, x) \end{aligned} \tag{10}$$

with

$$\zeta\left(T_i, \delta_i, \xi_i, \nu_i, t, x\right) = \frac{\mathbf{1}\left(T_i \leq t, \delta_i = 1\right)}{J\left(T_i^- \mid x\right)} - \int_0^t \left\{\mathbf{1}\left(T_i \geq v\right) + \mathbf{1}\left(T_i < v, \xi_i \nu_i = 1\right)\right\} \times \frac{dH^1\left(v \mid x\right)}{J^2\left(v^- \mid x\right)},$$

$$\widetilde{B}_{h_j i}\left(x\right) = \frac{1}{m\left(x\right)} \frac{1}{nh_j} K\left(\frac{x - X_i}{h_j}\right), \text{ for } j = 1, 2, \tag{11}$$

and $R_n\left(t, x\right)$ satisfies

$$\sup_{a \leq t \leq b, x \in I} \mid R_n\left(t, x\right) \mid = O\left((nh_1)^{-3/4} \left(\log n\right)^{3/4} + (nh_2)^{-3/4} \left(\log n\right)^{3/4}\right) \text{ a.s.} \tag{12}$$

Theorem 2 (Asymptotic normality) *Suppose that Assumptions 1–10 are satisfied, then, for $x \in I$ and $t \in [a, b]$ such that $\widehat{S}^c_{h_2}(t \mid x) > 1 - \widehat{p}^c_{h_1}(x)$, it follows that*

(i) If $nh_i^5 \frac{h_j}{h_1 + h_2} \to 0$, for $i, j = 1, 2$ $i \neq j$, then

$$\sqrt{\frac{nh_1h_2}{h_1 + h_2}} \left\{\widehat{S}^c_{0,h_1,h_2}\left(t \mid x\right) - S_0\left(t \mid x\right)\right\} \xrightarrow{d} N(0, s_c^2(t, x)),$$

where

$$s_c^2(t, x) = \begin{cases} s_{c,1}^2(t, x) & \text{if } \dfrac{h_1}{h_2} \to 0 \\ s_{c,2}^2(t, x) & \text{if } \dfrac{h_2}{h_1} \to 0 \\ \dfrac{C_2}{C_1 + C_2} s_{c,1}^2(t, x) + \dfrac{C_1}{C_1 + C_2}\left\{s_{c,2}^2(t, x) + 2s_{c,3}^2(t, x)\right\} & \text{if } \dfrac{h_1}{h_2} \to \dfrac{C_1}{C_2} \end{cases} \tag{13}$$

with $s_{c,1}^2(t, x)$, $s_{c,2}^2(t, x)$, $s_{c,3}^2(t, x)$ given in Equations (S.5)–(S.7) of Appendix B in the Supplementary Material, respectively, and C_1 and C_2 are constants.

(ii) If $nh_1^5 \to 0$ and $nh_2^5 \to C_2^5 > 0$, then

$$\sqrt{\frac{nh_1h_2}{h_1 + h_2}} \left\{\widehat{S}^c_{0,h_1,h_2}\left(t \mid x\right) - S_0\left(t \mid x\right)\right\} \xrightarrow{d} N(0, s_{c,1}^2(t, x)).$$

(iii) If $nh_1^5 \to C_1^5 > 0$ and $nh_2^5 \to 0$, then

$$\sqrt{\frac{nh_1h_2}{h_1 + h_2}} \left\{\widehat{S}^c_{0,h_1,h_2}\left(t \mid x\right) - S_0\left(t \mid x\right)\right\} \xrightarrow{d} N(0, s_{c,2}^2(t, x)).$$

(iv) If $nh_1^5 \to C_1^5 > 0$ *and* $nh_2^5 \to C_2^5 > 0$*, then*

$$\sqrt{\frac{nh_1h_2}{h_1+h_2}}\left\{\widehat{S}^c_{0,h_1,h_2}(t \mid x) - S_0(t \mid x)\right\} \xrightarrow{d} N(B_c(t,x), s_c^2(t,x))$$

where

$$B_c(t,x) = \sqrt{\frac{C_1C_2}{C_1+C_2}}\left\{C_1^2 B_{c,1}(t,x) + C_2^2 B_{c,2}(t,x)\right\},$$

with $B_{c,1}(t,x)$ *and* $B_{c,2}(t,x)$ *defined in Equations (S.3) and (S.4) of Appendix B in the Supplementary Material, and* $s_c^2(t,x)$ *is given in* (13).

The derivation of the asymptotic expressions of the bias, variance, along with the proofs for Theorems 1–2 are provided in Appendices B and C in the Supplementary Material, respectively.

2.4 Effect of ignoring the cure status information

In this section, we discuss the effect of ignoring the cure status information by comparing the dominant terms of the bias and variance of the proposed estimator in (6) with those of the nonparametric latency estimator proposed by López-Cheda et al. (2017b), which ignores the cure status:

$$\widehat{S}_{0,h}(t \mid x) = \frac{\widehat{S}_h(t \mid x) - \{1 - \widehat{p}_h(x)\}}{\widehat{p}_h(x)}, \tag{14}$$

where $\widehat{S}_h(t \mid x)$ is the generalized product-limit estimator of $S(t \mid x)$ by Beran (1981), and $1 - \widehat{p}_h(x)$ is the nonparametric cure probability estimator by Xu and Peng (2014), which ignores the observed cure status and considers cured individuals as simple censored observations. Hereafter, the estimator in (14), which depends on a single bandwidth h, will be referred to as LC.

Theorem 2 in López-Cheda et al. (2017b) presents asymptotic expressions for the bias and variance of the LC estimator, which are, respectively,

$$\begin{aligned}\mu_h(t,x) =& h^2\{B_1(t,x) + B_2(t,x)\} + O\left(h^4\right)\\ \sigma_h^2(t,x) =& \frac{1}{nh}\left\{s_1^2(t,x) + s_2^2(t,x) + 2s_3^2(t,x)\right\} + O(n^{-1}h),\end{aligned}$$

where $B_1(t,x)$ and $B_2(t,x)$ correspond to the terms $B_1^c(t,x)$ and $B_2^c(t,x)$ in Equations (S.3)–(S.4) of Appendix B in the Supplementary Material after replacing $G(t \mid x)$ with $G_0(t \mid x)$. As the bias of the proposed estimator depends on the derivatives of $1 - \pi(x)$ and $G_0(t \mid x)$, it is not straightforward to evaluate the exact gain of considering the cure status information in terms of bias.

On the other hand, the functions $s_1(t,x)$, $s_2(t,x)$ and $s_3(t,x)$ coincide with those of $s_{c,1}(t,x)$, $s_{c,2}(t,x)$ and $s_{c,3}(t,x)$ in Equations (S.5)–(S.7) of Appendix B in the Supplementary Material just replacing $H_{11}(t \mid x)$ with 0. When the cure information is included in the estimation, then $H^{11}(t \mid x) \geq 0$ and therefore $s^2_{c,i}(t,x) \leq s^2_i(t,x)$, for $i = 1, 2, 3$. So, ignoring the cure status information can increase the variance.

2.5 Bootstrap selection of the bandwidths

Bootstrap procedures for selecting the bandwidth for nonparametric estimation in the MCM have been studied in previous papers (López-Cheda et al. 2017a, b; Chown et al. 2019; Safari et al. 2022, 2021). Here we propose a bootstrap bandwidth selection method for the parameters (h_1, h_2) of the proposed estimator $\widehat{S}^c_{0,h_1,h_2}(t \mid x)$. The principle of the bootstrap-based selection methods is to choose the bandwidths h_1 and h_2 that minimize a bootstrap estimate of the mean integrated squared error (MISE). The bootstrap MISE can be approximated by

$$\text{MISE}^*_x(h_1,h_2) \simeq \frac{1}{B}\sum_{b=1}^{B}\int\left\{\widehat{S}^{c,*b}_{0,h_1,h_2}(v \mid x) - \widehat{S}^c_{0,g_{1_x},g_{2_x}}(v \mid x)\right\}^2 \omega(v,x)dv, \quad (15)$$

where $\widehat{S}^{c,*b}_{0,h_1,h_2}(t \mid x)$ is the estimator computed with the bth bootstrap resample and the bandwidths (h_1, h_2), and $\widehat{S}^c_{0,g_{1_x},g_{2_x}}(t \mid x)$ is the estimator computed with the original sample using the pilot bandwidths (g_{1_x}, g_{2_x}). Note that $\omega(v,x)$ is a nonnegative weight function, intended to give lower weight to the right tail of the distribution. Specifically, the steps to compute the bootstrap bandwidths for a fixed covariate value x are:

Step 1: With the original sample and the pilot bandwidths g_{1_x} and g_{2_x}, compute $\widehat{S}^c_{0,g_{1_x},g_{2_x}}(t \mid x)$ in (6).
Step 2: Choose two dense enough grids of bandwidths, $\{h_{11},\ldots,h_{1L}\}$ and $\{h_{21},\ldots,h_{2L}\}$.
Step 3: Generate B bootstrap resamples $\{(X_i^{(b)}, T_i^{*(b)}, \delta_i^{*(b)}, \xi_i^{*(b)}\nu_i^{*(b)}) : i = 1,\ldots,n\}$, for $b = 1,\ldots,B$.
Step 4: With the bth bootstrap resample and the bandwidths h_{1j} and h_{2k} compute $\widehat{S}^{c,*b}_{0,h_{1j},h_{2k}}(v \mid x)$, for $j,k = 1,\ldots,L$.
Step 5: For h_{1j} and h_{2k}, $j,k = 1,\ldots,L$, compute the $\text{MISE}^*_x(h_{1j},h_{2k})$ given by (15).
Step 6: The bootstrap bandwidths, h^*_{1x} and h^*_{2x}, are the bandwidths from the grids $\{h_{11},\ldots,h_{1L}\}$ and $\{h_{21},\ldots,h_{2L}\}$ that minimize $\text{MISE}^*_x(h_{1j},h_{2k})$ in (15).

The bootstrap resamples in Step 3 are generated as follows. Fix x, for $i = 1,\ldots,n$, set $X_i^{(b)} = X_i$ and generate a triplet $(T_i^{*(b)}, \delta_i^{*(b)}, \xi_i^{*(b)}\nu_i^{*(b)})$ from the weighted empirical distribution function

$$\widehat{F}_{g_{2x}}(t,d,v \mid x) = \sum_{i=1}^{n} B_{g_{2xi}}(x)\mathbf{1}\left(T_i \leq t, \delta_i \leq d, \xi_i\nu_i \leq v\right),$$

where $B_{g_{2x}}(x)$ are the Nadaraya-Watson weights in (4). We suggest the use of the bootstrap bandwidth selector in Safari et al. (2022) for the pilot bandwidth g_{1x}, and the bootstrap bandwidth selector in Safari et al. (2021) for g_{2x}.

This bootstrap resampling method is an adaptation of the simple weighted bootstrap in Li and Datta (2001) to the context of cured status partially known for some censored individuals. Without ties in the observed times, this simple weighted bootstrap and the corresponding obvious bootstrap resampling methods are equivalent (Safari 2022).

3 Simulation study

We conducted a simulation study to evaluate the finite sample performance of the estimator in (6). Particularly, we studied the effect of varying the proportion of individuals identified as cured, $\pi(x)$, the sample size, n, and the form of $p(x)$ and $S_0(t \mid x)$. The following estimators are considered for comparison: the nonparametric LC estimator in (14) that ignores the available cure status information and the semiparametric latency estimator, $S_0(t \mid x; \widehat{\beta})$, proposed by Bernhardt (2016), which includes the information of the identified cured individuals but does not benefit from the flexibility of the nonparametric estimator and strongly depends on the latency function to fit the assumed AFT model. Additional results on the effect of using either a single bandwidth h or two bandwidths (h_1, h_2) in the proposed estimator in (6) and the LC estimator in (14) are shown in Appendix D in the Supplementary Material.

The proportion of individuals identified as being cured was set to $\pi(x) = 0.2$ and 0.8. The censoring time C was generated, independently of Y given X, so that $C = \infty$ with probability $\pi(x)$, and with probability $1 - \pi(x)$ the value of C_0 was generated from a Weibull distribution with shape parameter $\alpha = 2$ and scale parameter $\beta = 2$. The covariate X was uniformly distributed on $[-20, 20]$. We assume four different models for the probability of the event $p(x)$: a logistic regression model $p_1(x) = (0.476 + 0.358x) / (1 + \exp(0.476 + 0.358x))$; the second and third models assume the constant functions $p_2(x) = 0.8$ and $p_3(x) = 0.2$; and the last one assumes a quadratic function $p_4(x) = 1 - 0.0025x^2$. Further, three different latency functions were defined as follows. The first latency function comes from an AFT model with lognormal distribution:

$$S_{0,1}(t \mid x) = \begin{cases} \dfrac{\Psi(4.305 + 0.15x) - \Psi(\log t - (0.3 - 0.15x))}{1 - \Psi(4.305 + 0.15x)} & \text{if } 0 \le t \le 4.605 \\ 0 & \text{otherwise,} \end{cases}$$

where $\Psi(\cdot)$ is the survival function of the standard normal distribution. This is the parametric model for the latency function that the semiparametric estimator $S_0(t \mid x, \widehat{\beta})$ assumes for the estimation of $S_0(t \mid x)$. The two other latency functions correspond to a truncated exponential function:

$$S_{0,2}(t \mid x) = \begin{cases} \dfrac{\exp(-\alpha(x)\, t) - \exp(-\alpha(x)\, 4.605)}{1 - \exp(-\alpha(x)\, 4.605)} & \text{if } 0 \le t \le 4.605 \\ 0 & \text{otherwise,} \end{cases}$$

Table 1 Characteristics of the simulated scenarios

Scenario	Setting		% Censoring		% cured
	$p(x)$	$S_0(t \mid x)$	$\pi(x) = 0.2$	$\pi(x) = 0.8$	
1	$p_1(x)$	$S_{0,1}(t \mid x)$	52.0	48.2	46.0
2	$p_2(x)$	$S_{0,1}(t \mid x)$	37.5	24.3	20.0
3	$p_3(x)$	$S_{0,1}(t \mid x)$	84.5	81.2	80.0
4	$p_1(x)$	$S_{0,2}(t \mid x)$	51.0	48.0	46.0
5	$p_4(x)$	$S_{0,3}(t)$	45.8	36.5	33.3

where $\alpha(x) = \exp((x + 20)/40)$, and

$$S_{0,3}(t) = \begin{cases} \dfrac{\exp(-t) - \exp(-4.605)}{1 - \exp(-4.605)} & \text{if } 0 \le t \le 4.605 \\ 0 & \text{otherwise.} \end{cases}$$

Notice that the latency functions are truncated at $\tau_0 = 4.605$ because the support of $G_0(t \mid x)$ has to be larger than that of $S_0(t \mid x)$. Table 1 shows the characteristics of the scenarios simulated in our study, which result from combining different models for $p(x)$ and $S_0(t \mid x)$. Scenarios 1–3 are intended to assess the effect of the form of $p(x)$ on the estimation of the latency function. Specifically, Scenario 1 considers the parametric functions assumed by the semiparametric estimator in Bernhardt (2016), so its performance is expected to be quite good. Note that Scenarios 2–5 serve to evaluate the performance of the semiparametric estimator when the parametric model assumptions are unsatisfied for either $p(x)$, $S_0(t \mid x)$ or both. Finally, in Scenario 5 the form of neither the probability of the event $p(x)$ nor the latency $S_0(t \mid x)$ match the models assumed by the semiparametric estimator.

We generated 1000 datasets for each scenario and sample sizes $n = 50$, 100 and 200. In addition, for the nonparametric estimators, the search of the optimal bandwidths (h_1, h_2) for the proposed estimator and h for the LC estimator, was performed in a grid of 51 values ranging from 5 to 100 and equispaced in logarithmic scale. The Epanechnikov kernel was used for nonparametric estimators. As weight function, we used $\omega(t, x) = \mathbf{1}(0 \le t \le w_x)$, where w_x is the 90th percentile of $S_0(t \mid x)$.

We approximated the integrated squared bias, integrated variance and mean integrated squared error for each estimator in every scenario. Figure 1 displays the MISE when $n = 100$ and $\pi(x) = 0.8$ for Scenarios 1–5. As expected, in Scenario 1 the semiparametric estimator behaves well for most values of the covariate X. The semiparametric estimator also gives acceptable results in Scenarios 2 and 3 because of the AFT model in $S_{0,1}(t \mid x)$ and the cure probability function is constant. Nevertheless, in these scenarios, the proposed estimator is highly competitive and beats the semiparametric estimator for a wide range of values of the covariate. The proposed estimator outperforms the semiparametric estimator in Scenarios 4–5 because the underlying logistic and/or AFT model assumptions for the cure probability and the latency are violated. In most scenarios, it is observed that the proposed estimator outperforms the LC estimator.

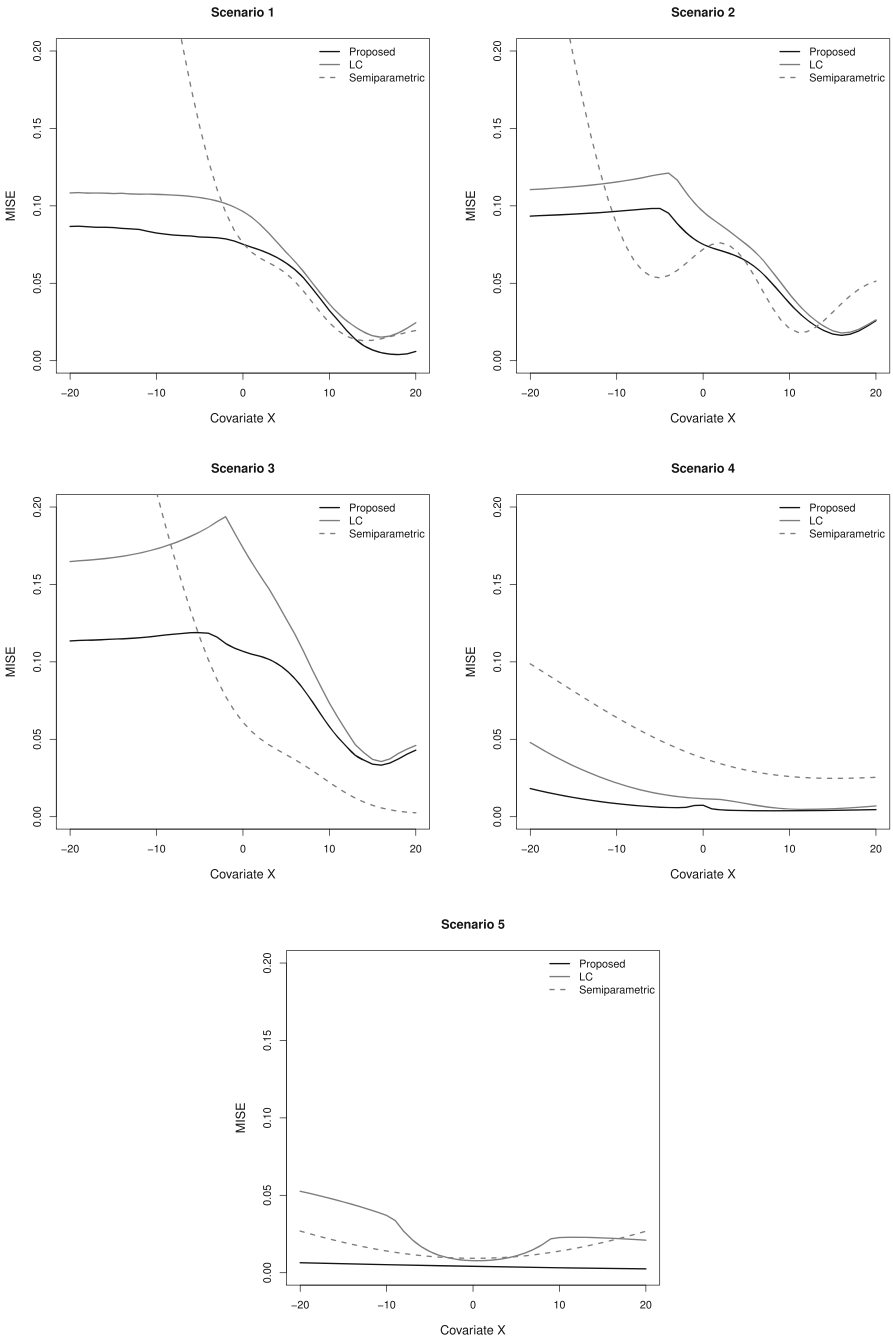

Fig. 1 MISE of the proposed, LC and semiparametric estimators (the first two, computed with the optimal bandwidths) in Scenarios 1–5 for $n = 100$ and $\pi(x) = 0.8$

Table 2 provides information on the MISE, the integrated squared bias and integrated variance of the proposed estimator, the LC estimator, and the semiparametric estimator for $x = -10, 0$ and 10 with $\pi(x) = 0.8$ and $n = 100$. This table shows that the estimator $\widehat{S}^c_{0,h_1,h_2}(t \mid x)$ substantially outperforms the LC estimator in all scenarios.

The impact of varying n and $\pi(x)$ was assessed in Scenario 4, and the results are presented in Appendix D in the Supplementary Material. As expected, we observe that increasing the sample size leads to a decrease in the MISE values of all the estimators (see Table S1 and Figure S1 in the Supplementary Material). The proposed estimator consistently possessed the minimum MISE.

The performance of the bootstrap bandwidth selector was assessed in Scenario 4 using a total of 1000 samples, for $\pi(x) = 0.8$ and $n = 100$, $B = 500$ bootstrap resamples and a grid of 51 bandwidths equispaced in a logarithmic scale, from $h_{i1} = 5$ to $h_{i51} = 100$, $i = 1, 2$. Figure 2 shows the contour plots of the MISE and the density of the selected bootstrap bandwidths $h^*_{1_x}, h^*_{2_x}$ for the covariate values $x = -2, 2, 8$ and 14. For this sample size and this scenario, the bootstrap bandwidth selector seems to perform well.

4 Real data analysis

During pandemic waves the demand for hospital resources, such as hospital beds, increases dramatically. Accurate estimates of the trajectory of a patient and the lengths of stay (LoS) from one hospital facility (ward, ICU) to another, are crucial for healthcare authorities to suitably plan capacity. The total LoS of a patient is divided into several stages according to the patient pathway, and each time is modeled separately (Currie et al 2020; García-Vicuña, 2020).

For illustration purposes, we consider one specific length of stay, the time of a patient in hospital ward until admission to ICU. We illustrate the performance of the proposed estimator in (6) to model the distribution of that LoS given age and sex as covariates of interest. In this context, the event of interest is admission to ICU from hospital ward. The estimation is carried out with the hospitalized COVID-19 patients in Galicia, Spain, during the first wave (March 6 – May 5, 2020); see also Gude-Sampedro et al. (2021), López-Cheda et al. (2021a), and Safari et al. (2022) for other results on this dataset. During that period, nearly 2,380 patients were hospitalized for at least one day, 197 (8.3%) of them needing ICU care. This caused the Galician Healthcare Service (2023) to experience a significant increment of the demand of hospital beds, ICU beds, and other health care resources.

The patients in the dataset are divided into three groups: 197 (8.3%) of them who experienced the event (required ICU) during the period of study; 1,966 (82.6%) who are acknowledged to be cured from the event, they are not requiring ICU either because they were discharged alive (1,638 patients) or died (328 patients); and 217 (9.12%) patients still in hospital ward by the end of the study (censored observations). The percentage of patients known to be cured from ICU admission is very high (82.6%), so the non-inclusion of that information in the estimation procedure is expected to give inefficient and possibly biased estimates. The median of the observed LoS in the

Table 2 Integrated squared bias (Ibias^2), integrated variance (Ivar) and MISE of the proposed estimator, LC estimator, both computed with the optimal MISE bandwidths, and the semiparametric estimator, in the simulated scenarios for $x = -10, 0$ and 10 with $\pi(x) = 0.8$ and $n = 100$

Scenario	x	Proposed				LC				Semiparametric		
		(h_1, h_2)	Ibias^2 $\times 10^{-2}$	Ivar $\times 10^{-2}$	MISE $\times 10^{-2}$	h	Ibias^2 $\times 10^{-2}$	Ivar $\times 10^{-2}$	MISE $\times 10^{-2}$	Ibias^2 $\times 10^{-2}$	Ivar $\times 10^{-2}$	MISE $\times 10^{-2}$
1	−10	(14.70, 15.61)	4.76	3.49	8.24	22.36	8.42	2.32	10.74	27.72	2.30	30.02
	0	(10.26, 8.08)	5.53	2.00	7.52	11.57	7.94	1.70	9.64	6.64	0.94	7.58
	10	(15.61, 6.75)	2.47	0.77	3.24	5.99	2.94	0.72	3.66	2.18	0.27	2.45
2	−10	(28.42, 30.17)	9.05	0.59	9.65	36.11	10.98	0.56	11.54	8.32	0.47	8.79
	0	(8.07, 8.57)	6.21	1.30	7.51	8.08	8.16	1.46	9.62	6.82	0.38	7.20
	10	(6.35, 6.75)	2.93	0.77	3.70	5.64	3.24	1.06	4.30	1.49	0.59	2.08
3	−10	(74.11, 32.03)	9.29	2.38	11.67	100	14.27	3.03	17.30	0.21	1.20	1.41
	0	(13.85, 13.85)	7.29	3.39	10.68	13.04	12.94	4.43	17.37	0.24	0.69	0.93
	10	(10.26, 10.90)	3.55	2.27	5.82	10.26	4.41	2.93	7.34	0.25	1.14	1.39
4	−10	(100, 36.11)	0.29	0.55	0.85	74.11	0.61	1.57	2.18	3.20	3.21	6.41
	0	(15.61, 21.06)	0.15	0.59	0.74	100	0.06	1.10	1.16	2.76	1.02	3.78
	10	(58.32, 54.93)	0.00	0.38	0.39	12.28	0.03	0.47	0.50	2.21	0.39	2.60
5	−10	(83.55, 94.18)	0.01	0.52	0.52	25.21	2.85	0.86	3.70	0.21	1.20	1.40
	0	(88.71, 100)	0.01	0.40	0.41	10.26	0.14	0.64	0.78	0.25	0.69	0.93
	10	(88.71, 100)	0.01	0.32	0.32	25.21	1.77	0.50	2.27	0.25	1.14	1.40

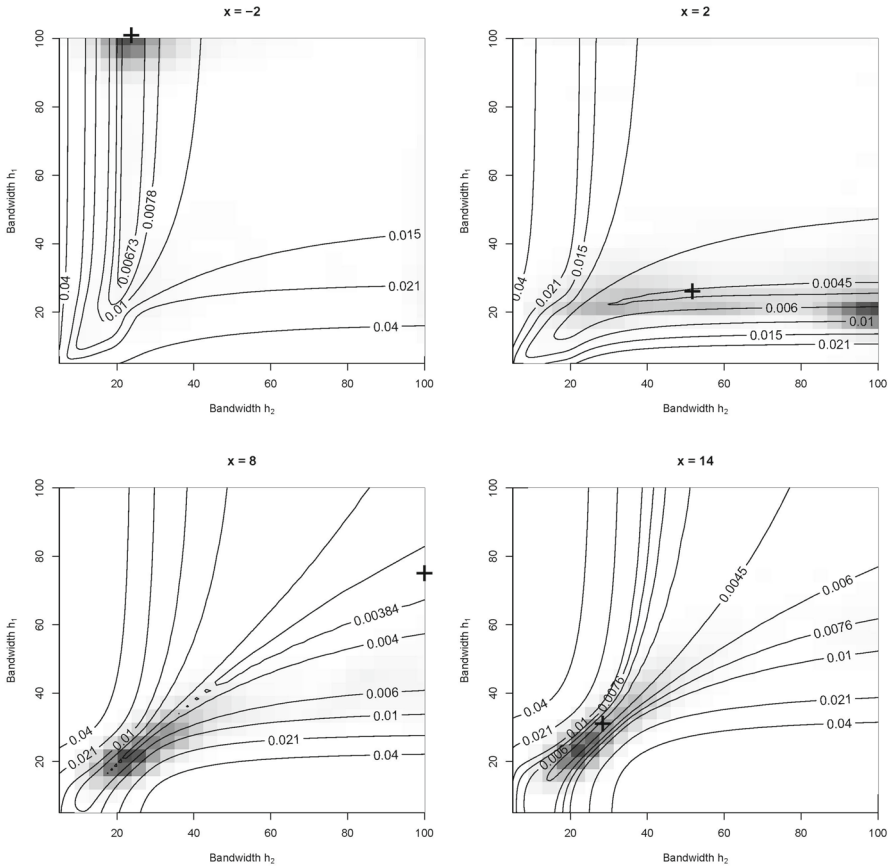

Fig. 2 Contour plots of the MISE of the proposed estimator in Scenario 4, for $x = -2, 2, 8$ and 14 with $\pi(x) = 0.8$ and $n = 100$, as a function of the two bandwidths, h_1, h_2. The optimal bandwidth where the minimum MISE is reached is marked with a cross. The joint density of the bootstrap bandwidths $(h^*_{1,x}, h^*_{2,x})$ is also shown in shades of grey, where a darker grey represents a greater density

hospital ward for patients admitted to the ICU was 3 days (interquartile range (IQR): $2-5$). The median of the observed LoS in the hospital ward for patients who were discharged alive was larger, 9 days (IQR: $6-14$), than those who died in the hospital ward, 7 days (IQR: $4-12$). In this example, the value of Y was observed (uncensored) not only for patients with bad health status such that were admitted to the ICU within the follow-up time ($C_i > Y_i$), but also for patients with bad health status such that they died before entering ICU ($C_i = \infty$), also those with good health status and discharged alive without needing ICU ($C_i = \infty$). Only patients who were still in the hospital ward, with unknown prognosis and future pathway, were censored ($C_i < Y_i$) at the end of the study. Consequently, the distribution of C does not appear to necessarily depend on the health status of the patient, thus the independence assumption between Y and C conditional on X might be plausible. Moreover, to apply our estimator we

Table 3 Basic characteristics of COVID-19 patients hospitalized in Galicia from March 6 to May 5, 2020 ($n = 2{,}380$)

Variable	Count (%)	Uncensored/uncured	Censored		
		ICU admission	Cured†		Cure unknown
			Dead	Discharged	
Age					
0 – 24	22 (0.9)	0	0	17	5
25 – 54	359 (15.1)	25	5	316	13
55 – 64	354 (14.9)	41	24	271	18
65 – 74	582 (24.5)	96	35	411	40
75 – 84	571 (24.0)	34	101	377	59
85+	492 (20.7)	1	163	246	82
Sex					
Female	1,118 (47.0)	55	136	822	105
Male	1,262 (53.0)	142	192	816	112

† "Cured" from the event "admission to ICU"

assume that condition (8) applies, as suggested by the test of Maller and Zhou (1992) (p-value < 0.001).

Table 3 describes the COVID-19 data by age and sex. The proportion of patients being admitted to the hospital ward ranged from 0.9% for 0−24 years old patients to 24.5% for the 65−74 years group. Patients aged between 65 and 74 years were the most prevalent in ICU admission (96 patients), while none among the youngest age group was admitted to the ICU. In this data, 47% were females and 53% males. Besides, men were more likely to be admitted to the ICU compared to women (Safari et al. 2022).

The survival curve of the LoS until admission to ICU was estimated separately for males and females with the proposed unconditional estimator $\widehat{S}_n^c(t)$ of the latency in (7), the semiparametric estimator in Bernhardt (2016) and the unconditional version of the LC estimator, that ignores that a high percentage of patients is not requiring ICU. The conditional survival curve was estimated also for some fixed ages, using the proposed estimator in (6) and the bootstrap bandwidth selector in Sect. 2.5, the semiparametric estimator in Bernhardt (2016), and the LC estimator with the bootstrap bandwidths computed with the `R` package `npcure` (López-Cheda et al. 2021b). The results are displayed in Fig. 3. We see that the survival curves using the proposed estimator and the semiparametric estimator are closer to each other, which suggests that the semiparametric AFT model might be appropriate for modelling the LoS until admission to ICU. Meanwhile, the survival curve estimated using the LC estimator appears to be too optimistic, which may indicate that ignoring the information that many patients are not experiencing the event could yield extremely biased estimates.

As can be seen in Fig. 3, patients aged 58 and 74 years spent the same median time (3 days) in hospital ward before admission to ICU. By sex, the median time

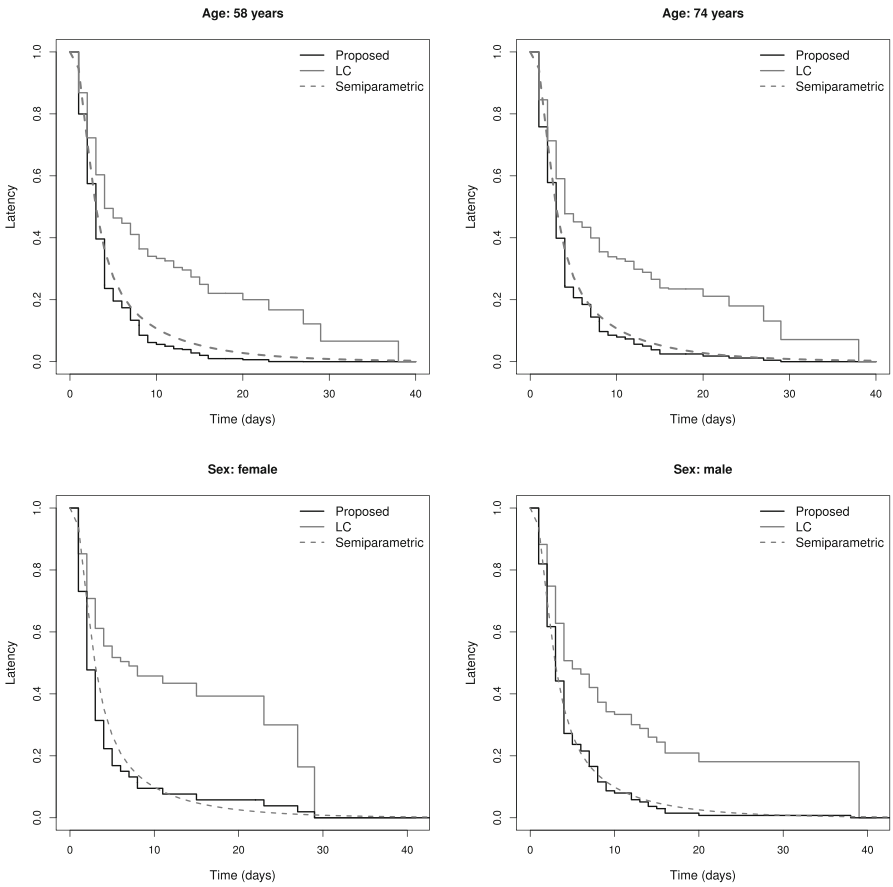

Fig. 3 Latency estimates of the time in hospital ward until admission to ICU of COVID-19 patients requiring ICU aged 58 and 74 years obtained using the proposed estimator $\widehat{S}^c_{0,h_1,h_2}(t \mid x)$, the LC estimator $\widehat{S}_{0,h}(t \mid x)$ (both computed using the bootstrap bandwidths), and the semiparametric estimator. Also shown are the latency estimates by sex, both computed using the proposed estimator $\widehat{S}^c_{0,n}(t)$, the estimator that ignores the known cured observations $\widehat{S}_{0,n}(t)$ and the semiparametric estimator

before ICU admission was also the same (3 days). These estimates were obtained using $\widehat{S}^c_{0,h_1,h_2}(t \mid x)$.

It is worth mentioning that the proposed estimator can be applied to estimate the distribution of the LoS of patients who die in the hospital before being admitted to ICU (event), just considering patients discharged or admitted to ICU as cured individuals. Similarly, the distribution of the LoS of patients who are discharged without ICU admission (event) can be estimated using the proposed methodology just by noticing that patients who die in the hospital before being admitted to ICU or patients admitted to ICU can be considered as cured. R code for implementation and reproduction of the results in this section is available on GitHub (see Appendix E in the Supplementary Material).

5 Discussion

In this paper we developed a completely nonparametric approach to estimate the latency function in the MCM when the cure status information is available for a subset of the censored observations. The main advantage of the proposed estimator is that, unlike other alternatives in the literature, it can handle any type of covariates without the need of assuming any (semi)parametric model, which provides flexibility to the method. The asymptotic properties of the estimator were given, and a procedure for the selection of the bandwidths was provided, showing its nice performance in practice. Furthermore, the computational time of the estimator was considerably fast, while bootstrap bandwidth method was relatively time-consuming.

The effect of including the cure status information in the estimation procedure was demonstrated to be clearly beneficial by comparing the proposed estimator with the nonparametric estimator by López-Cheda et al. (2017b). We proved that including the known cures in the estimation reduces the variance asymptotically. The gain in terms of bias is not straightforward from a theoretical point of view, as it depends on the conditional probability of knowing the cure status information and the censoring distribution, although the simulation results show great improvement in terms of bias for all the scenarios.

Approaches to include the cure information in the estimation of the MCM are scarcely developed in the literature. Many of them are based on a cure threshold, in which an individual is assumed to be cured from the event when the observed time to event surpasses a threshold, that is, the event was not observed for a fixed period time (see, e.g., Laska and Meisner 1992; Nieto-Baraja and Yin 2008). The proposed methodology not only encompasses this setup, but also generalizes it, as it can be applied when a subject is considered cured based on external information, that is, when the "cured" times are random.

Our proposed method could be extended in several ways. The covariates in our study are time-independent but in biometrical studies time-dependent covariates might easily arise. In addition, the proposed estimator is tailored to handle one single covariate. It is well-known that the behavior of nonparametric smooth estimators quickly deteriorates as the dimension of the covariate vector increases, see Stone (1980). To avoid the curse of dimensionality when estimating the latency function with multiple covariates, alternative techniques such as product kernels or a single-index model, may be more appropriate. Finally the proposed methodology is aimed at the MCM, but other cure models such as the promotion time cure model (Tsodikov 1998) can be applied to analyze survival data with a cure rate. All of these additional complications require extensions of our methodology, and we leave these issues as possibilities for future work.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s10985-023-09591-x.

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